6T SRAM Cell Design and Analysis

(Using GPDK 180 nm Technology)

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Tools: Cadence Virtuoso

This project was independently designed and analyzed as a personal learning initiative.

Abstract

This project focuses on the **design and functional analysis of a 6-transistor (6T) SRAM cell** using the **GPDK 180 nm CMOS technology** in **Cadence Virtuoso**. The SRAM cell is designed at the transistor level and verified for correct operation in **hold, read, and write modes** for both logic '0' and logic '1'. Proper **transistor sizing** is employed to ensure reliable read stability and write ability. Functional correctness is validated through transient simulations, while advanced stability metrics such as **Static Noise Margin (SNM)** are planned for future analysis.

1. Introduction

Static Random Access Memory (SRAM) is widely used in modern digital systems due to its **high speed, low latency, and no refresh requirement**. The 6T SRAM cell is the most commonly used SRAM architecture, consisting of two cross-coupled inverters and two access transistors. This project aims to understand the behavior of a 6T SRAM cell by designing and analyzing it using a standard CMOS process.

2. Technology and Design Specifications

2.1 Technology Used

- **Process**: GPDK 180 nm CMOS
- **Design Level**: Transistor level
- **Tool**: Cadence Virtuoso

2.2 Transistor Dimensions

Transistor Type	Width (W)
Access NMOS (WA)	2 µm
Pull-up PMOS (WP)	2.4 µm
Pull-down NMOS (WN)	3 µm

3. 6T SRAM Cell Architecture

The 6T SRAM cell consists of:

- Two **pull-up PMOS transistors**
- Two **pull-down NMOS transistors**
- Two **access NMOS transistors**

The cross-coupled inverter structure enables data storage, while the access transistors connect the cell to the bit lines during read and write operations.

4. Transistor Sizing Justification

4.1 Cell Ratio (CR)

$$\text{Cell Ratio (CR)} = \frac{W_{pull-down}}{W_{access}}$$

$$CR = \frac{3\ \mu m}{2\ \mu m} = 1.5$$

Justification:

- A CR greater than 1 ensures that the **pull-down NMOS is stronger than the access NMOS**.
- This prevents accidental flipping of the stored data during a read operation.
- A CR of 1.5 provides adequate **read stability** in 180 nm technology.

4.2 Pull-up Ratio (PR)

$$\text{Pull-up Ratio (PR)} = \frac{W_{pull-up}}{W_{access}}$$

$$PR = \frac{2.4\ \mu m}{2\ \mu m} = 1.2$$

Justification:

- A moderate PR ensures that the **access NMOS can overpower the pull-up PMOS during write**.
- This improves write ability without significantly increasing leakage.
- A PR close to 1 is commonly used for balanced write performance.

5. Modes of Operation

5.1 Hold Mode

- **WL = 0**
- Access transistors are OFF
- Cross-coupled inverters retain stored data
- Verified for both logic ‘0’ and logic ‘1’

Observation:
The stored value remains stable with no voltage degradation, confirming correct hold operation.

5.2 Write Operation

Write ‘0’

- BL = 0, BLB = VDD
- WL = 1
- Internal node is successfully pulled low

Write ‘1’

- BL = VDD, BLB = 0
- WL = 1
- Internal node transitions to high

Observation:
The access NMOS successfully overcomes the pull-up PMOS, enabling reliable write for both logic states.

5.3 Read Operation

- BL and BLB are held at defined logic levels prior to read
- WL = 1
- One bit line exhibits a voltage change based on the initialized internal node logic state

Observation:
Read operation does not disturb the stored value, indicating sufficient read stability due to proper CR selection.

6. Simulation Results

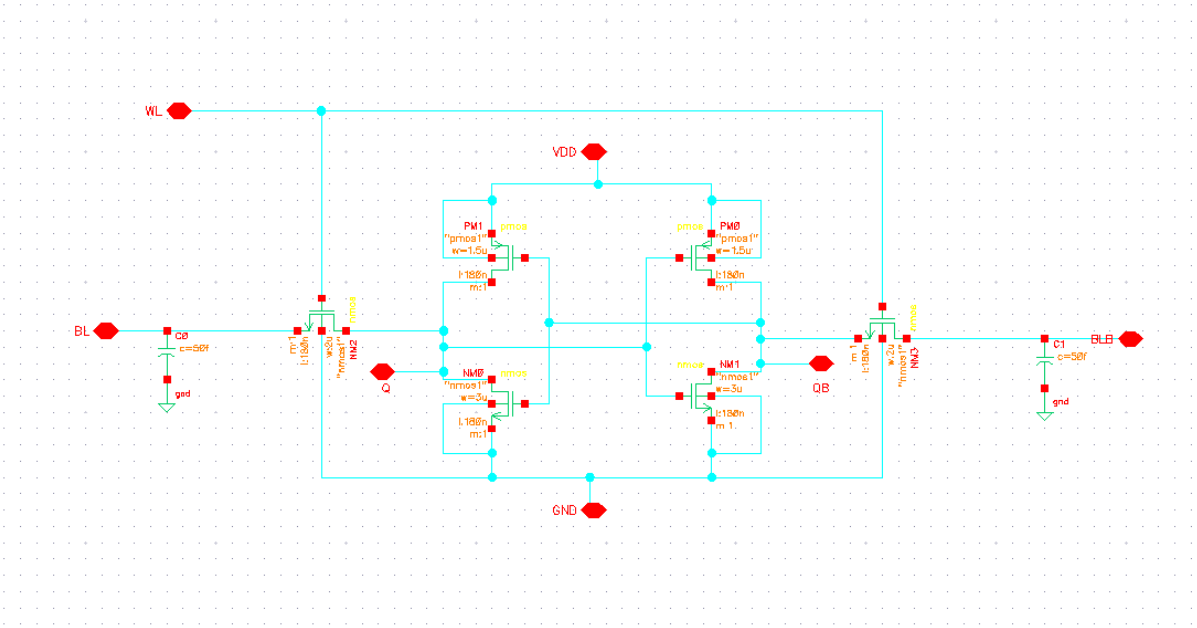


Figure 6.1: Transistor-Level Schematic of 6T SRAM Cell (GPDK 180 nm)

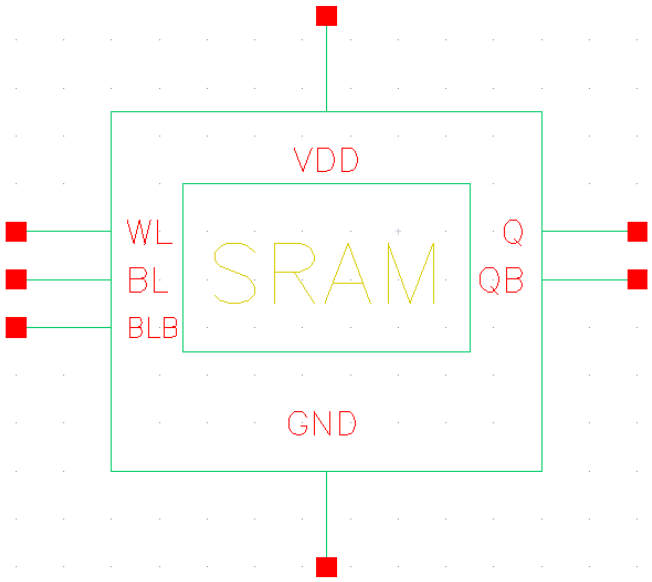


Figure 6.2: Symbol View of the 6T SRAM Cell

Figures 6.1 and 6.2 show the transistor-level schematic and the corresponding symbol view of the designed 6T SRAM cell, respectively. The schematic represents the complete cell implementation using GPDK 180 nm technology, while the symbol view is used for testbench-based functional verification. Together, they enable efficient simulation and validation of correct hold, read, and write operations for both logic '0' and logic '1'.

Read ‘0’ operation:

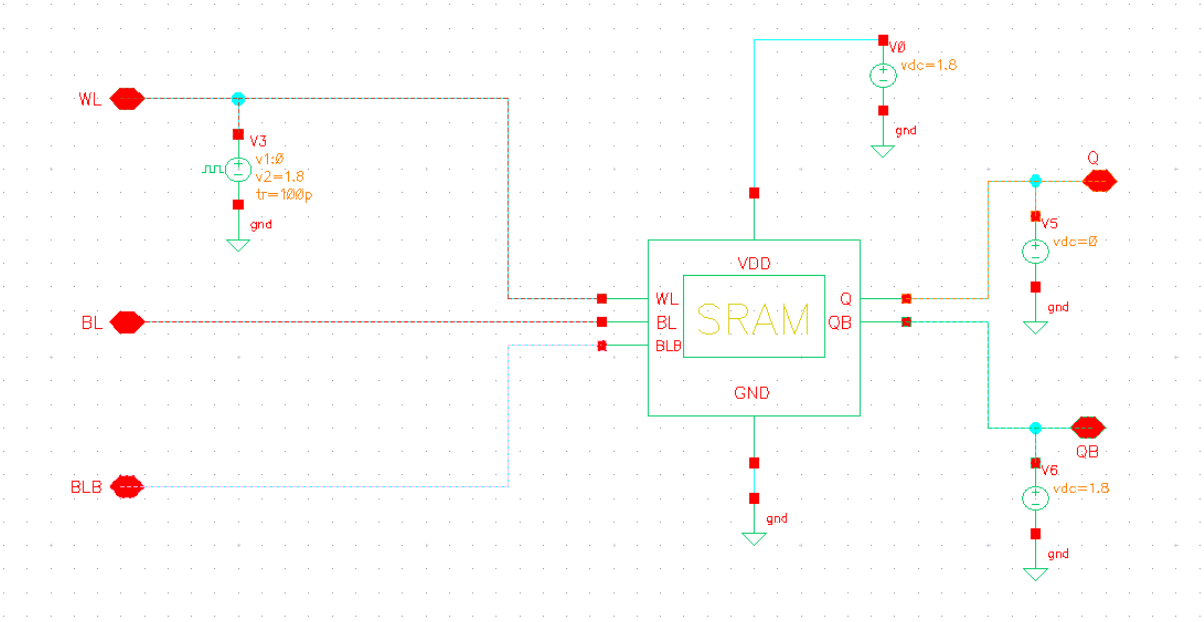


Figure 6.3: Testbench circuit configuration for read ‘0’ operation of the 6T SRAM cell.

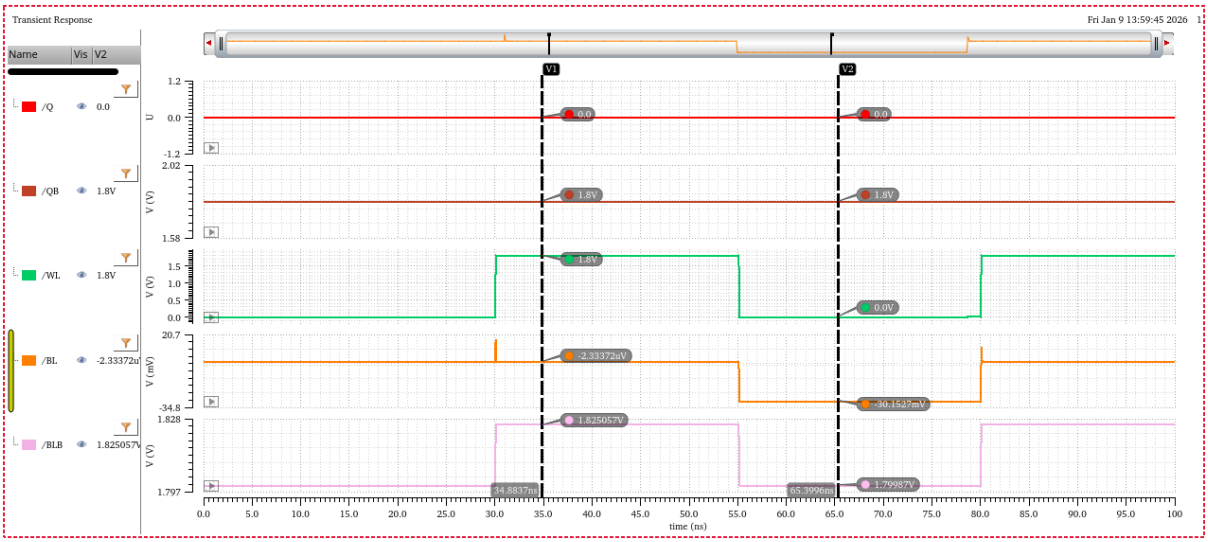


Figure 6.4: Transient response waveform of the 6T SRAM cell during read ‘0’ operation showing WL, BL, BLB, Q, and QB signals.

The read ‘0’ operation was verified by initializing the internal nodes with Q at logic ‘0’ and QB at logic ‘1’, followed by word line activation while observing the bit-line behavior. As observed from the waveform, when WL is enabled, the bit line connected to the internal node at logic ‘0’ exhibits a slight discharge, while the complementary bit line remains close to VDD. The internal nodes Q and QB remain stable throughout the read operation, indicating that the initialized cell state is not disturbed. This confirms adequate read stability of the SRAM cell for logic ‘0’ with the chosen transistor sizing.

Read ‘1’ operation:

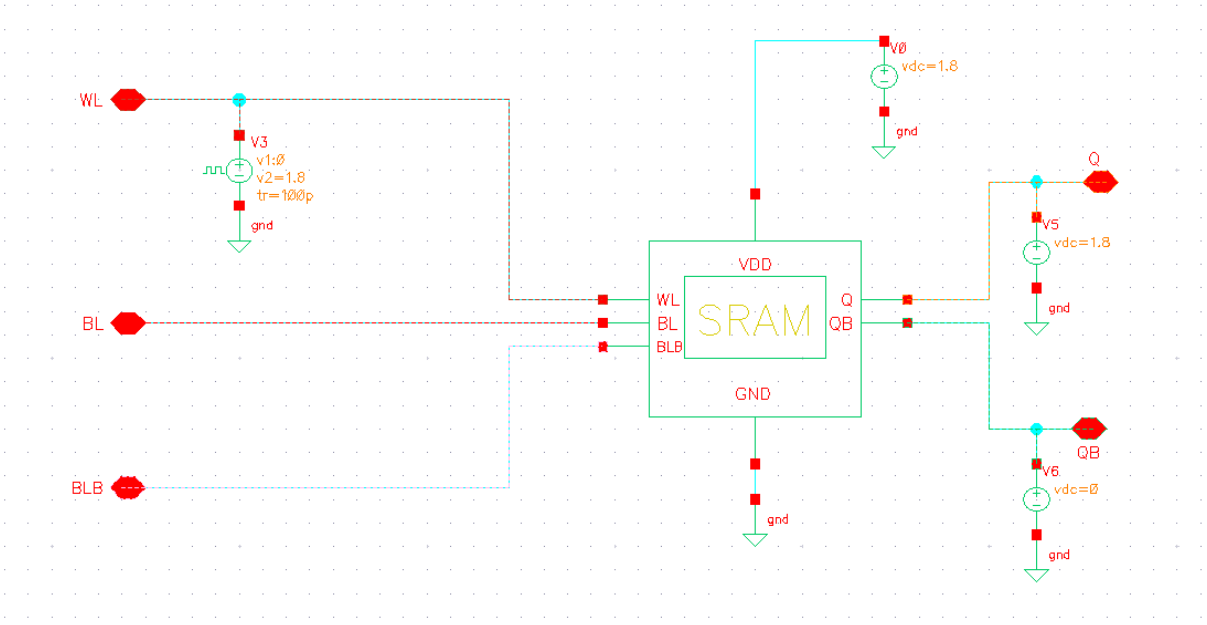


Figure 6.5: Testbench circuit configuration for read ‘1’ operation of the 6T SRAM cell.

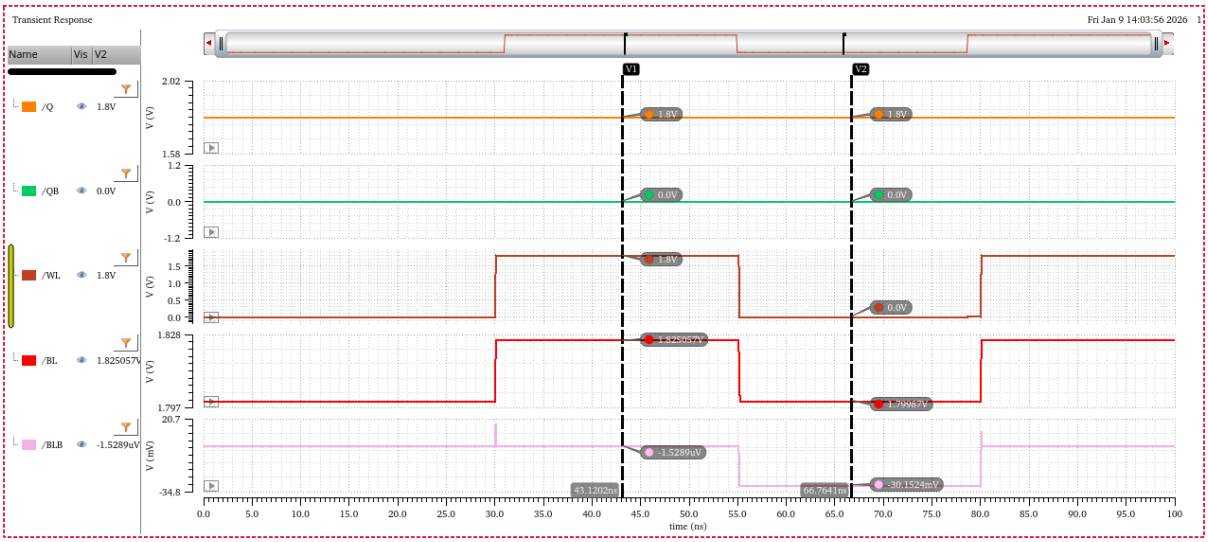


Figure 6.6: Transient response waveform of the 6T SRAM cell during read ‘1’ operation showing WL, BL, BLB, Q, and QB signals.

For the read ‘1’ operation, the internal nodes were initialized with Q at logic ‘1’ and QB at logic ‘0’, followed by word line activation while monitoring the bit-line response. The simulation results show a voltage drop on the bit line connected to the internal node at logic ‘0’, while the opposite bit line stays near VDD. The internal nodes retain their original logic levels during the entire read cycle, demonstrating non-destructive read behavior. This validates correct and stable read operation for logic ‘1’ under the implemented design conditions.

Write ‘0’ and Write ‘1’ Operations:

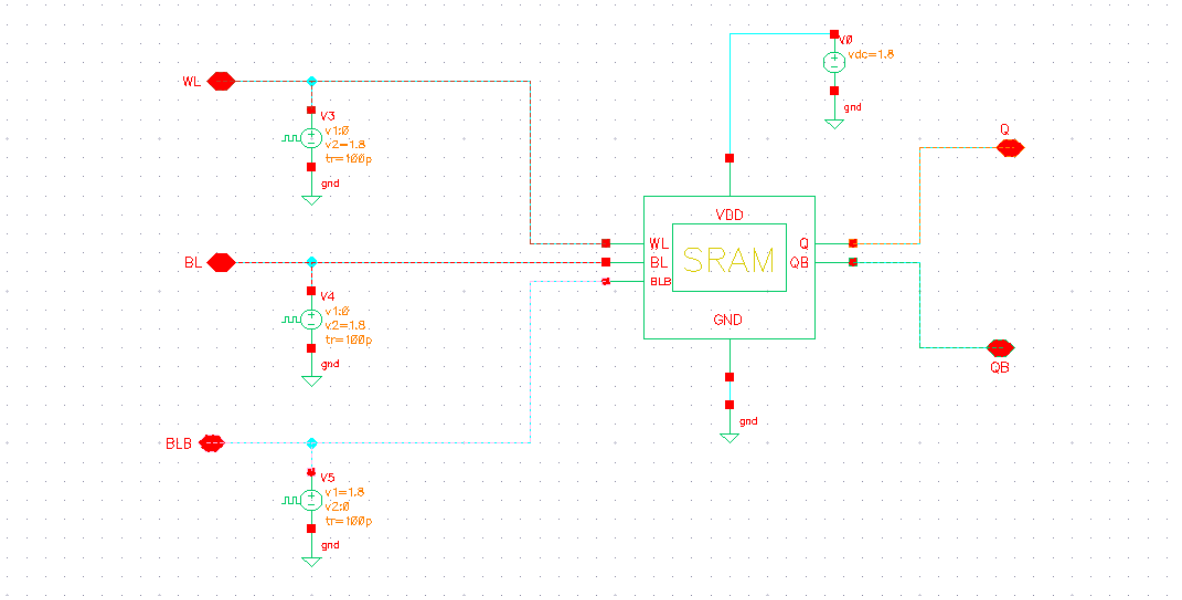


Figure 6.7: Testbench circuit configuration for write ‘0’ and write ‘1’ operations of the 6T SRAM cell.

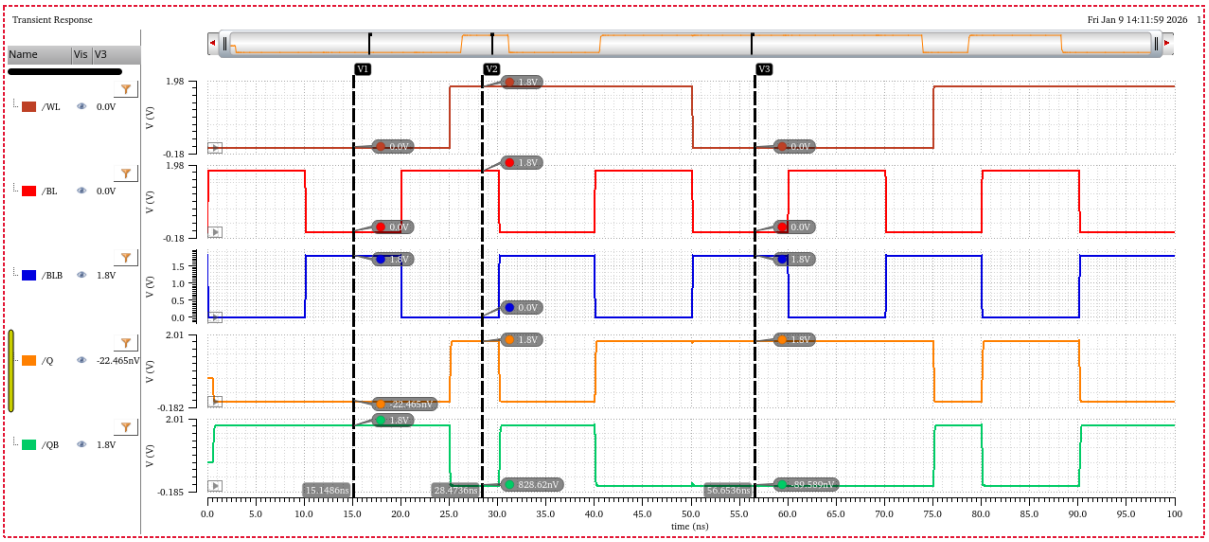


Figure 6.8: Transient response waveform of the 6T SRAM cell during write ‘0’ and write ‘1’ operations showing WL, BL, BLB, Q, and QB signals.

The write operations were verified by driving complementary data on the bit lines while enabling the word line. During write ‘0’, BL is driven low and BLB high, forcing the internal node Q to logic ‘0’, while write ‘1’ is achieved by reversing the bit-line inputs. The waveforms confirm successful overwriting of the previously stored data, with Q and QB switching to the intended logic levels within the active word line period. This demonstrates adequate write ability of the SRAM cell with the selected transistor sizing.

Hold ‘0’ and Hold ‘1’ Operations:

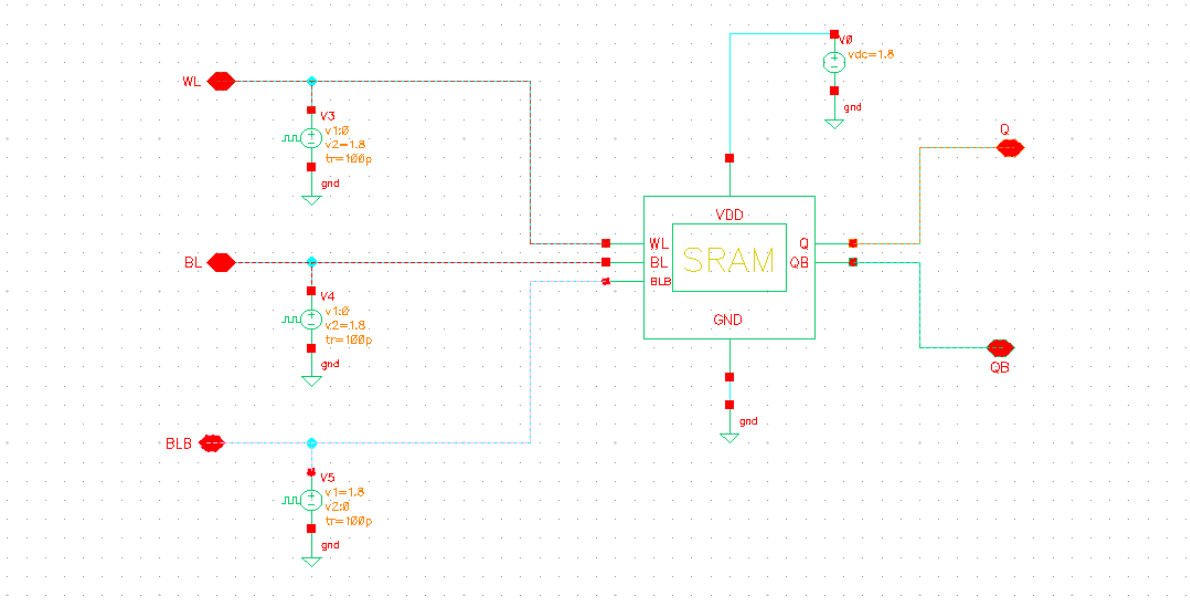


Figure 6.9: Testbench circuit configuration for hold ‘0’ and hold ‘1’ operations of the 6T SRAM cell.

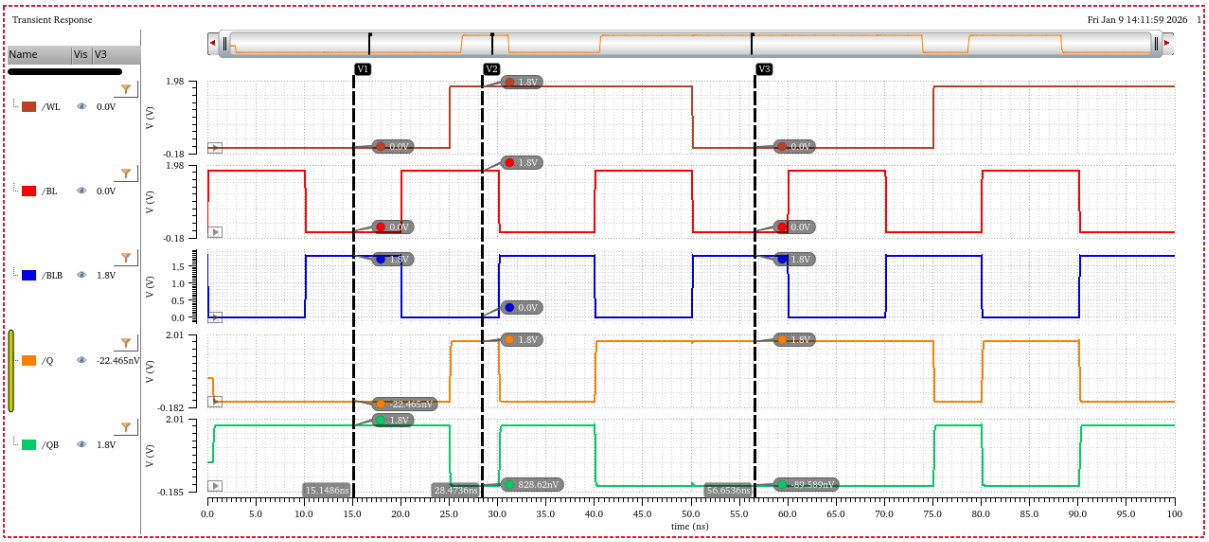


Figure 6.10: Transient response waveform of the 6T SRAM cell during hold ‘0’ and hold ‘1’ operations showing WL, BL, BLB, Q, and QB signals.

Hold mode operation was evaluated by keeping the word line disabled while maintaining the initialized internal node states as either logic ‘0’ or logic ‘1’. The simulation results show that the internal nodes Q and QB retain their respective logic levels without degradation, independent of bit-line activity. This confirms the stability of the cross-coupled inverter structure and validates correct data retention in hold mode.

- Correct functional behavior was observed for **hold, write, and read operations** for both logic ‘0’ and logic ‘1’.
- No unintended data flipping occurred during read operations, indicating adequate read stability.
- Simulation waveforms were clean, consistent, and repeatable across multiple cycles.

7. Discussion

- Higher pull-down strength improves read stability
- Balanced pull-up sizing allows reliable write operation
- Proper transistor sizing is critical for robust SRAM operation

8. Future Work

- Static Noise Margin (SNM) extraction using butterfly curves
- SRAM array implementation

9. Conclusion

A 6T SRAM cell was successfully designed and analyzed using **GPDK 180 nm technology** in Cadence Virtuoso. Functional verification for **hold, read, and write operations** was completed for both logic states. The chosen transistor sizing achieved a balanced trade-off between read stability and write ability.